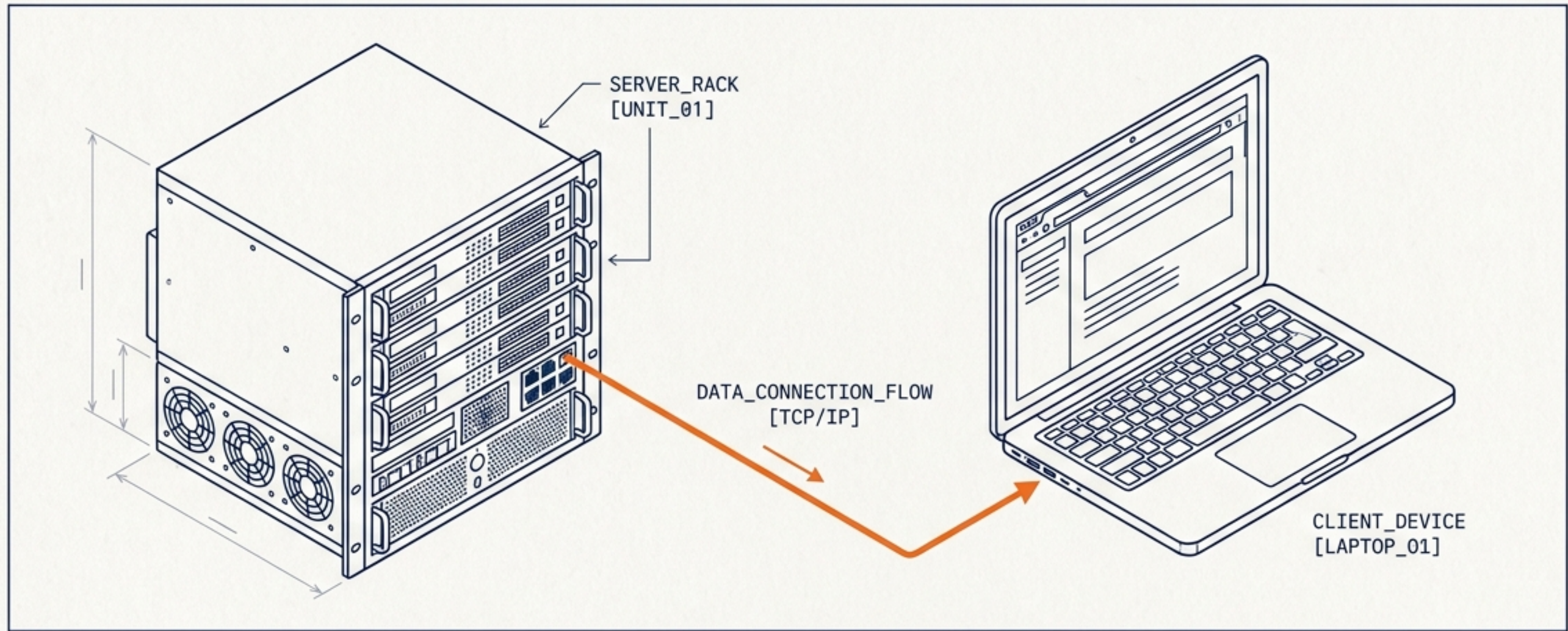




The Architecture of the Web

From Static Pages to Social Systems: A Blueprint of the WWW

The Interface

1

A hypertext-based system providing a uniform, user-friendly interface to access Internet resources.

The Information Space

2

An interconnected web of items of interest (resources).

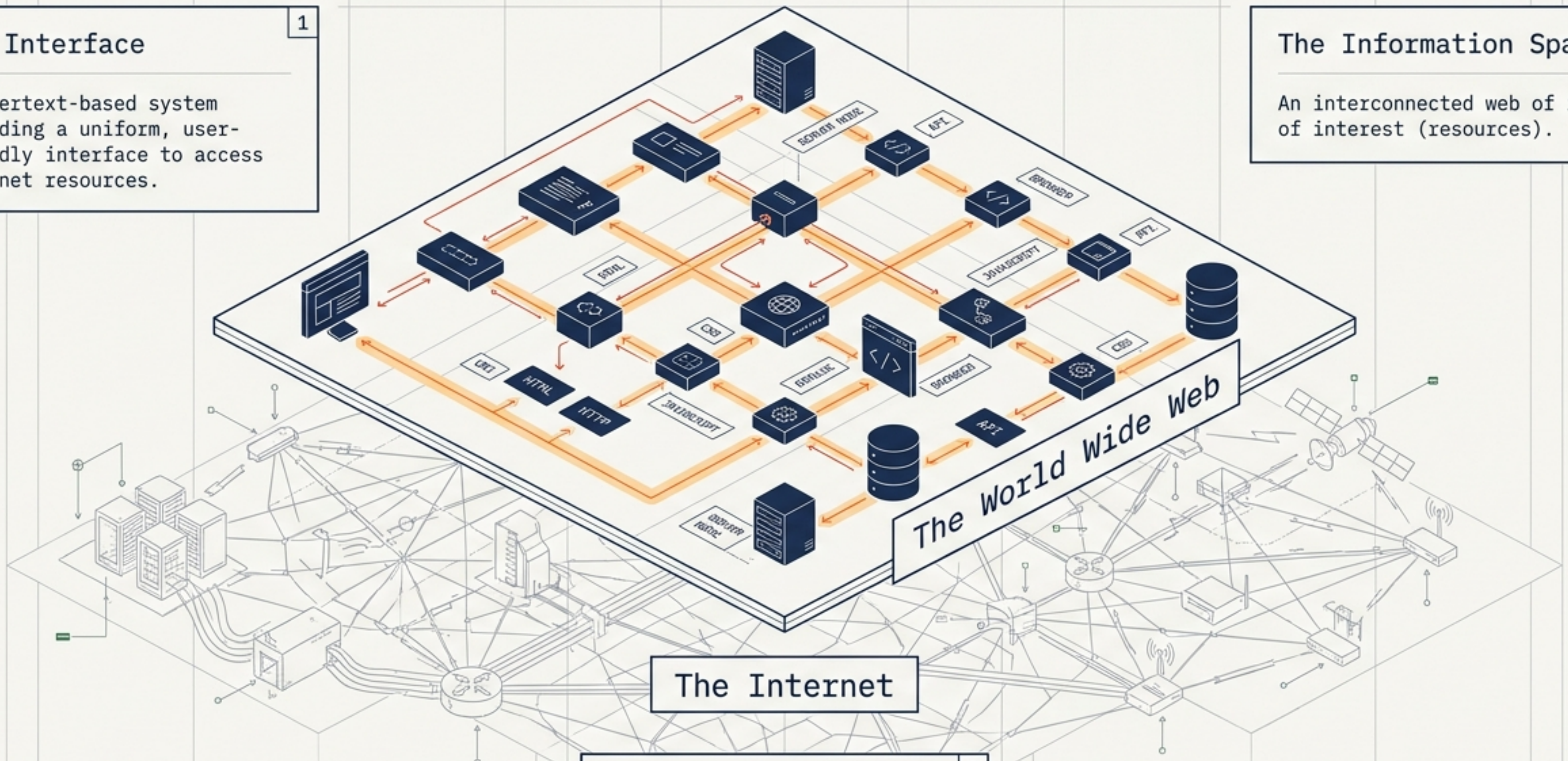
The Internet

The Address Book

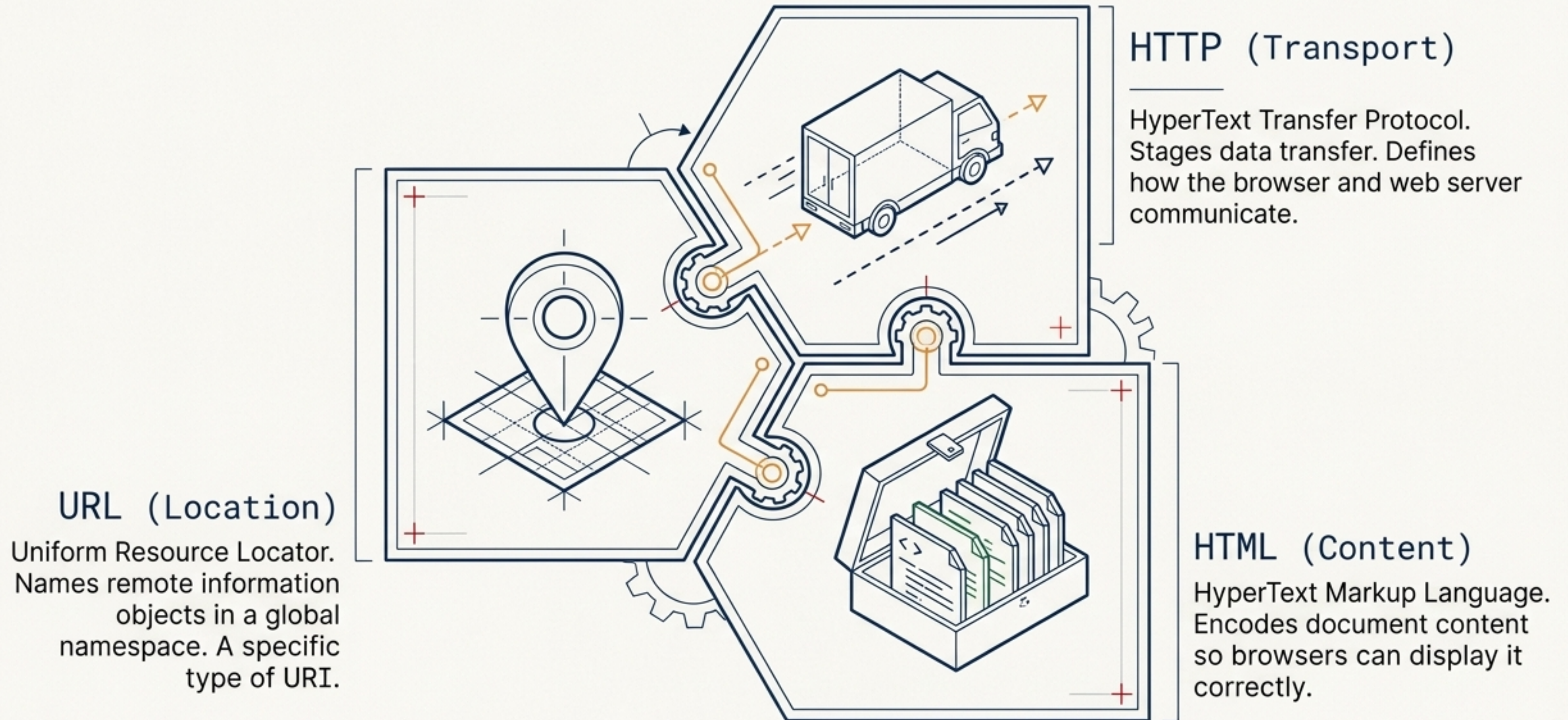
3

Every resource is identified by a global identifier known as a URI (Uniform Resource Identifier).

The World Wide Web

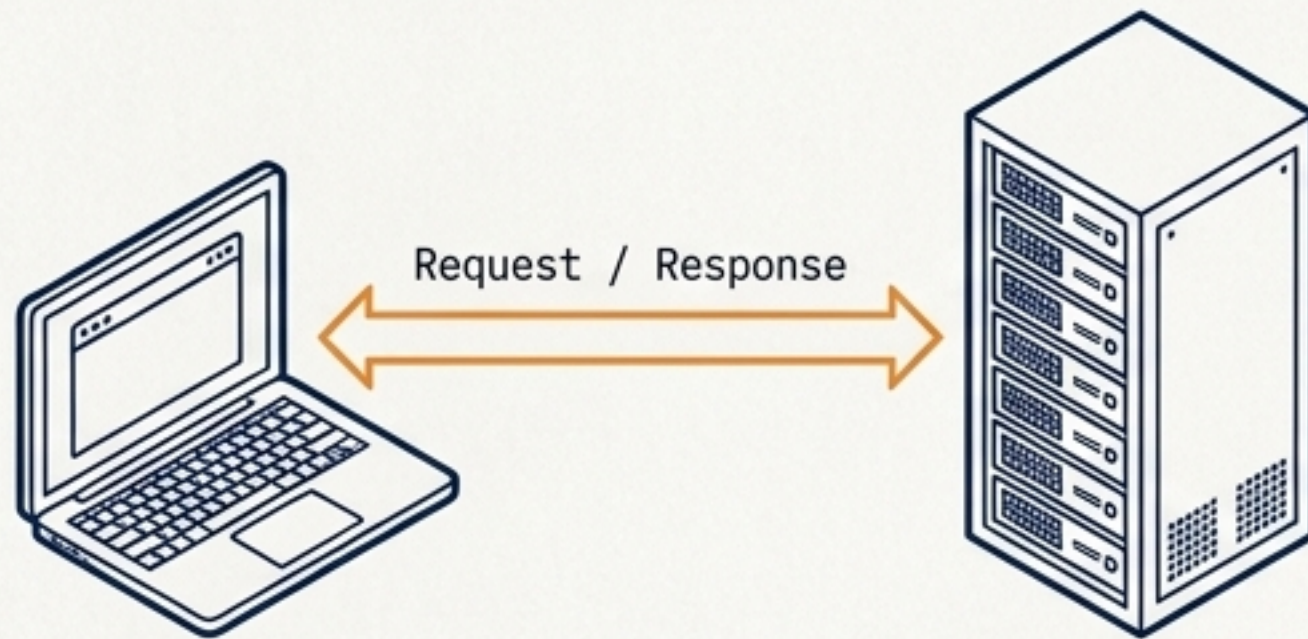


Three Key Standards of the WWW



Architectural Topologies

Two-Tier Architecture

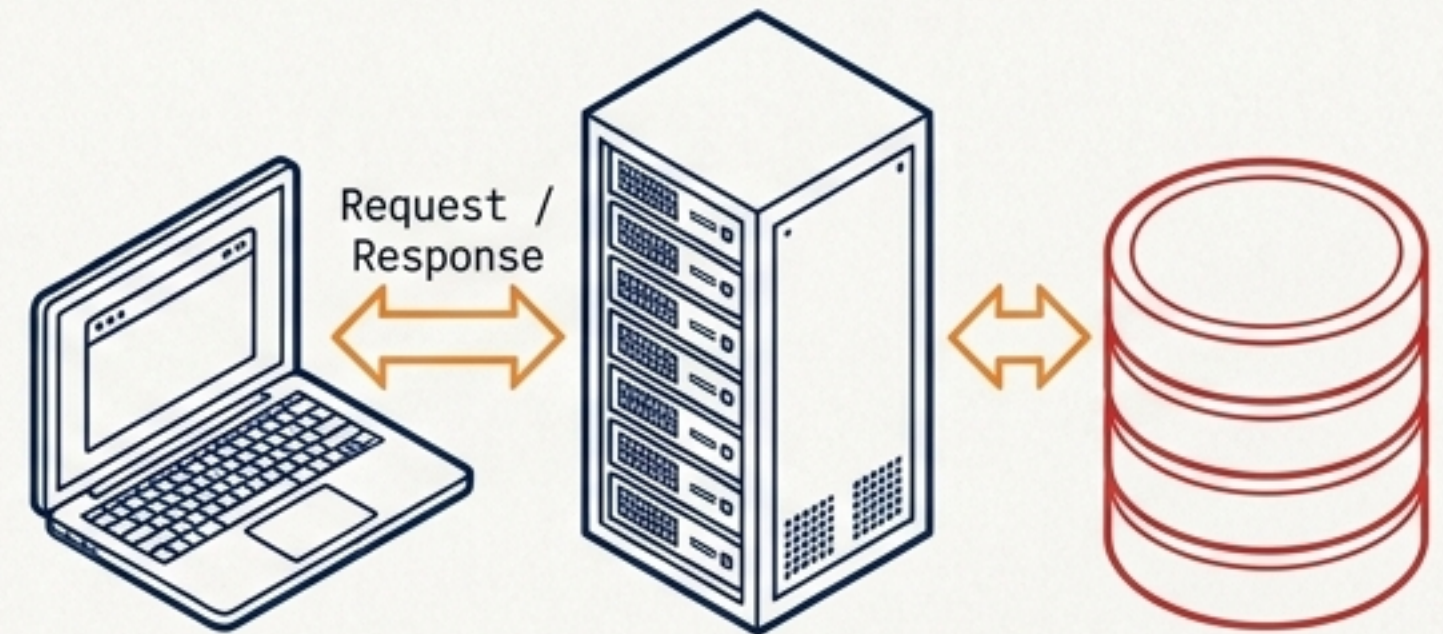


Client /
Web Browser

Server /
Web Server

The fundamental architecture of the WWW. The client requests a page; the server retrieves and returns it.

Three-Tier Architecture



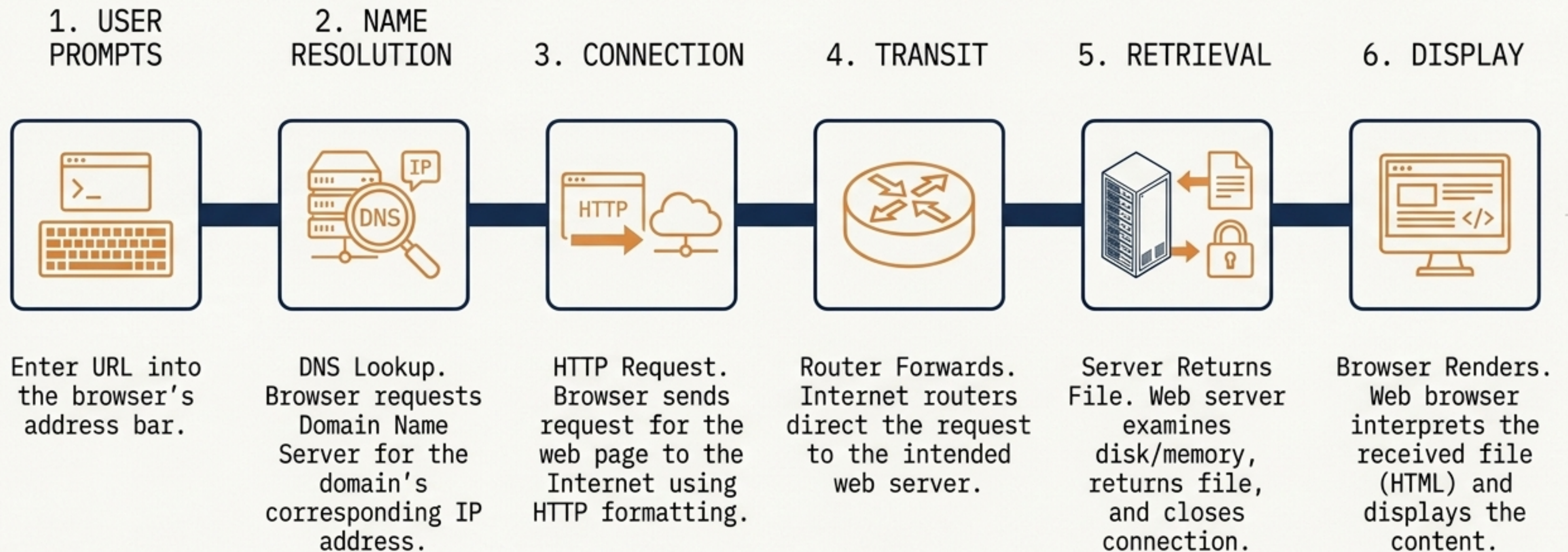
Client

App/Web
Server

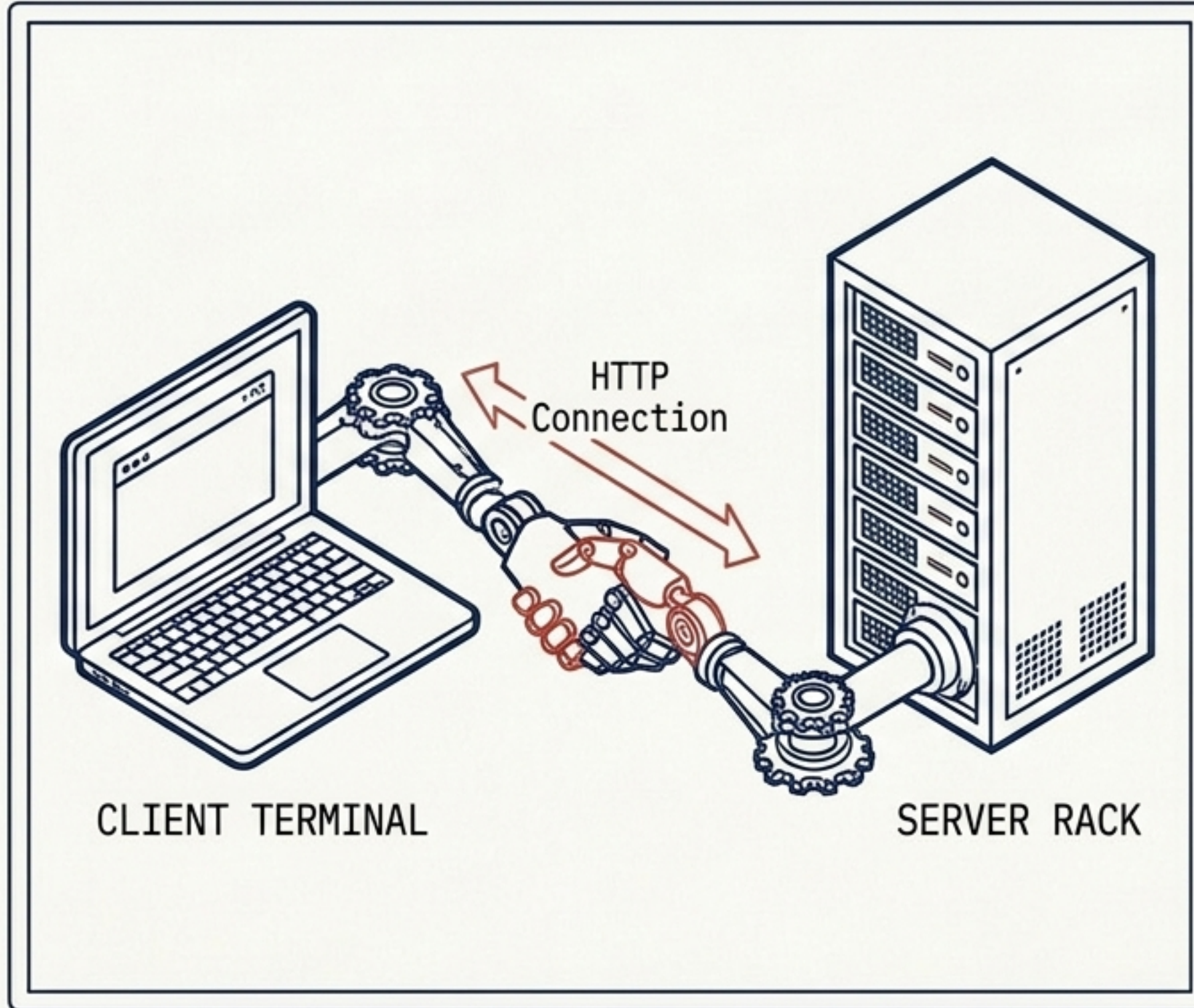
Database /
Data Tier

Activated when web pages interact with a database, adding a dedicated data tier to the request flow.

The Journey of a Request



The Rules of Transfer: HTTP



HTTP is an Internet standard that dictates how messages are formatted and transmitted.

Command Structure

Defines exact actions web servers and browsers must take in response to commands.

The Engine at Work

When a user enters a URL, the browser actually sends an HTTP command directing the server to fetch and transmit the requested page.

The Evolution of the Web

Web 1.0

The Read-Only Web

A simple content-delivery system.

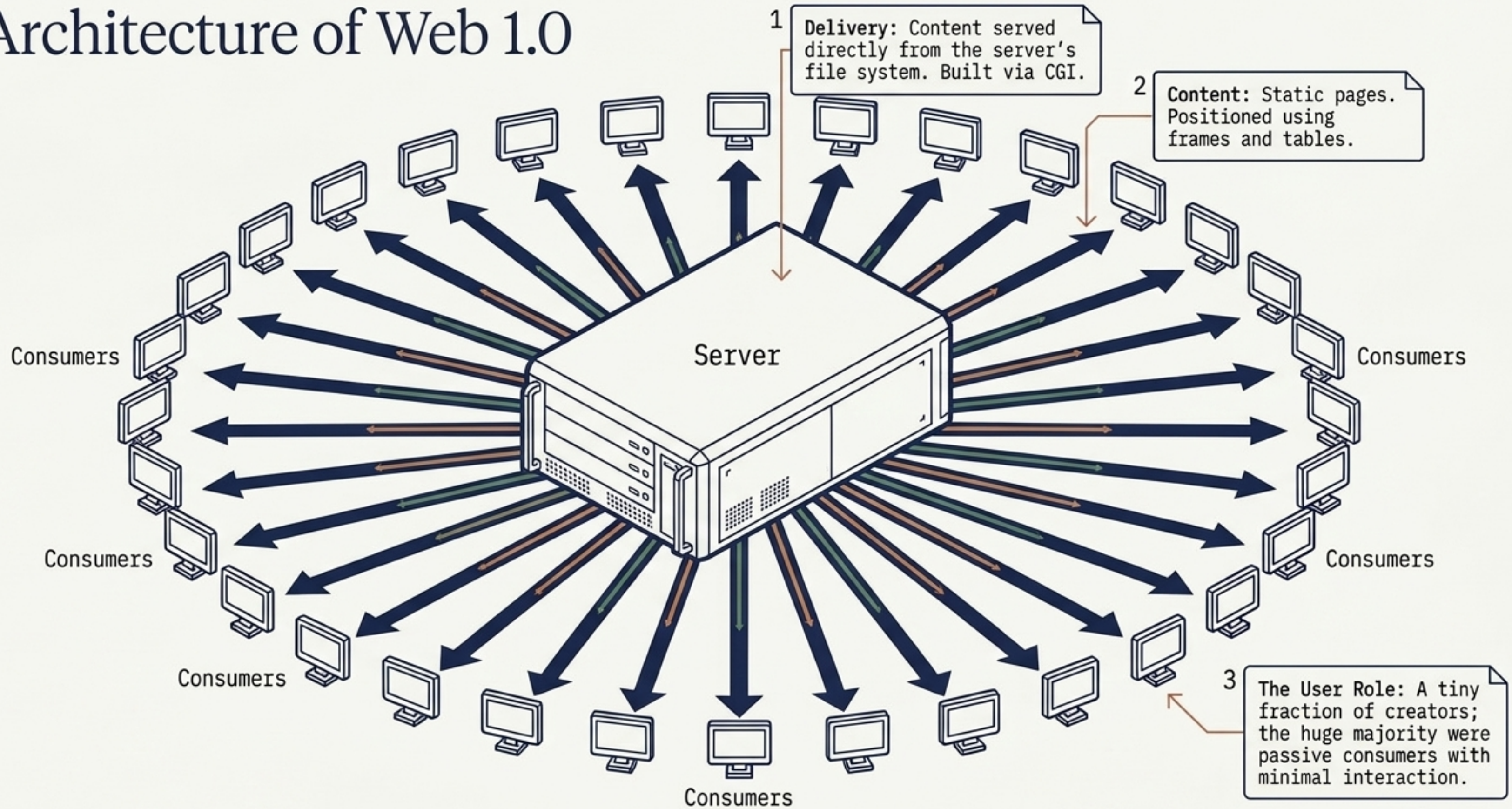
(Tim Berners-Lee)

Web 2.0

The Participative Social Web

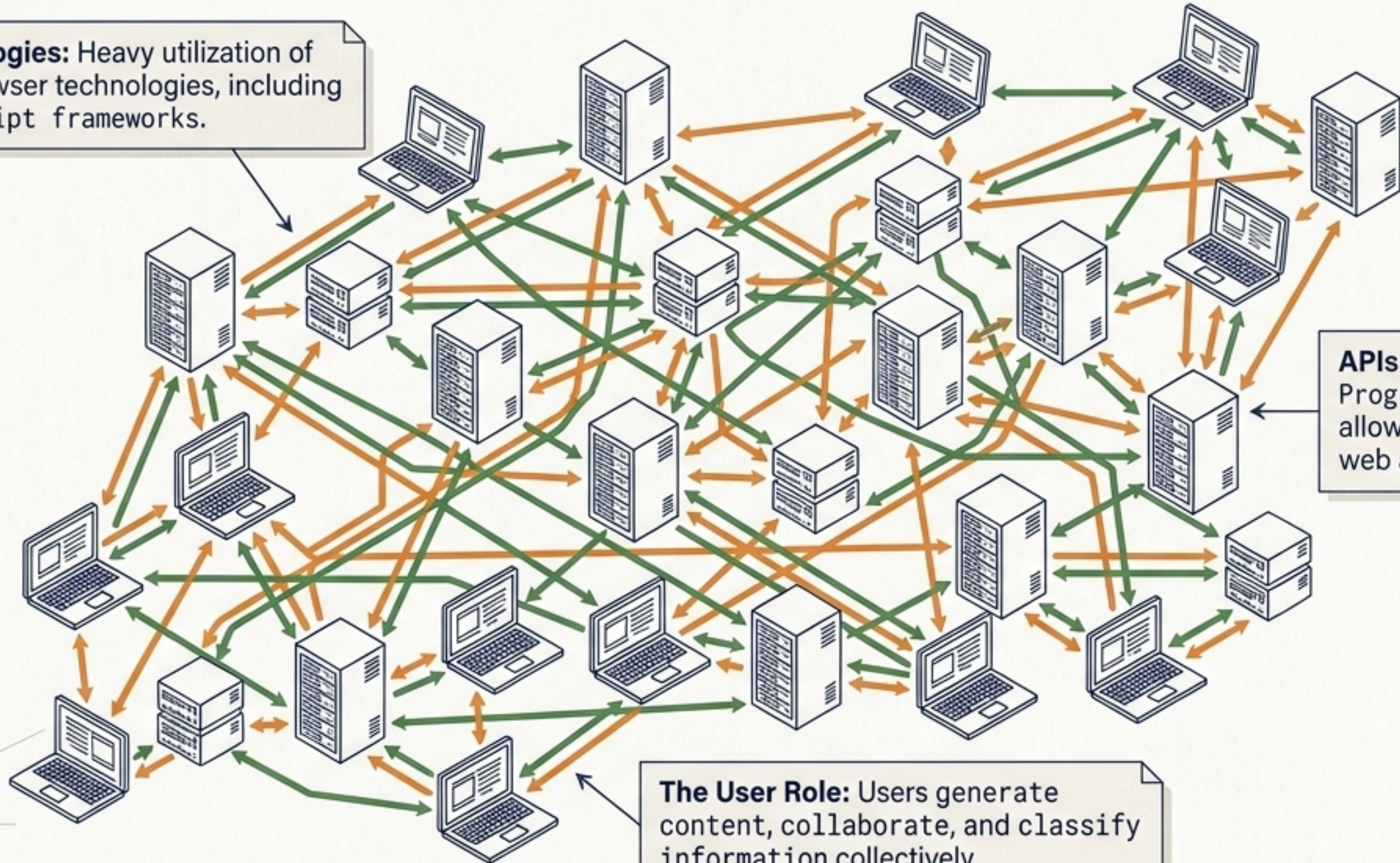
A highly interactive, collaborative platform. This evolution was not a change in technical specifications, but a cumulative shift in software design and end-user behavior.

Architecture of Web 1.0



Architecture of Web 2.0

Technologies: Heavy utilization of new browser technologies, including JavaScript frameworks.



APIs: Developed Application Programming Interfaces allow software to treat the web as a software platform.

The User Role: Users generate content, collaborate, and classify information collectively.

The Paradigm Shift Matrix

	Web 1.0	Web 2.0
Nature	Read-only	Read-write / Participative
Content State	Mostly static pages	Dynamic, responsive to input
User Role	Consumers of content	Creators & collaborators
Interaction Level	Very limited	High (Comments, Sharing, APIs)
Core Focus	Publishing & delivery	Collaboration & community

The Structural Requirement of Web 2.0

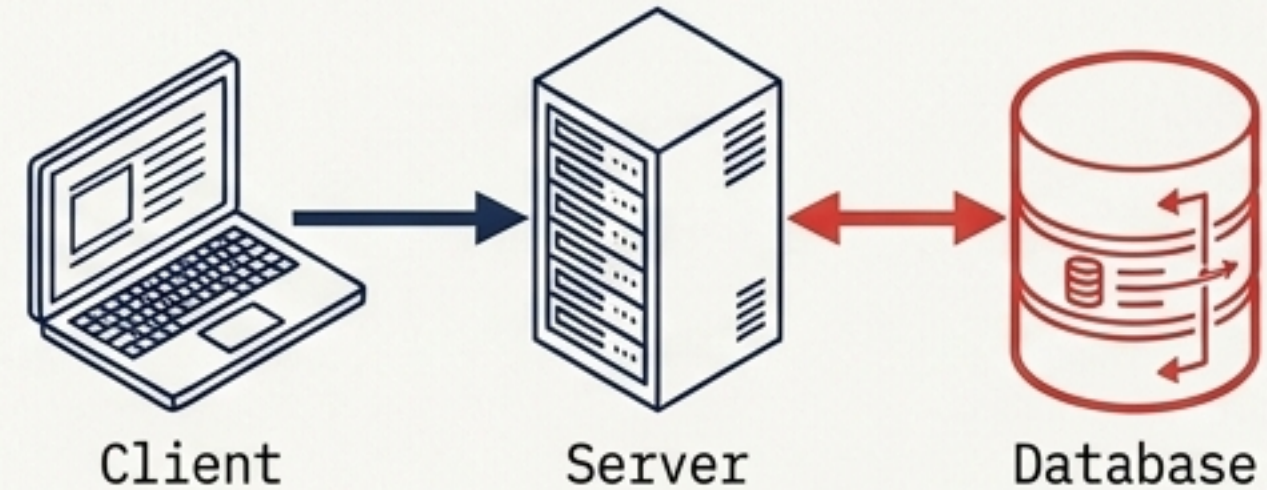
Web 1.0 (Static Publishing)



2-Tier Architecture

Static publishing relied on direct Client-to-Server file fetching.

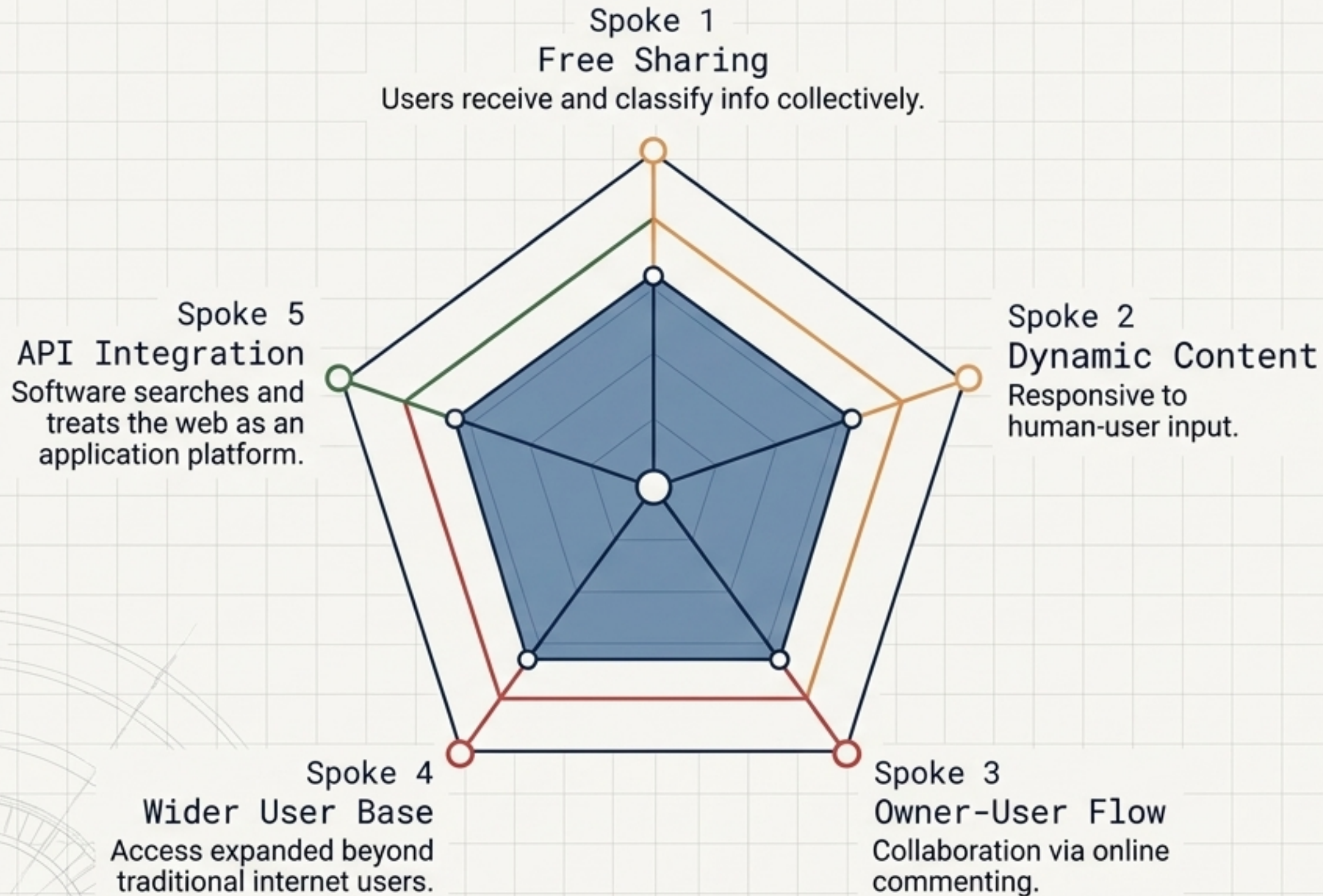
Web 2.0 (Social & Participative)



3-Tier Architecture

Insight: The participative web of user-generated content and dynamic APIs fundamentally requires the Data Tier (Databases).

Major Features of Web 2.0



Common Applications



Podcasting



Blogging



Tagging



Social
Networking



Social
Bookmarking

Schematic Summary

The Engine (Architecture)

- **WWW** = Hypertext interface via **URIs**.
- **2-Tier** = Client + Server (Base).
- **3-Tier** = Client + Server + Database (Dynamic).

The Standards (Protocols)

- **HTML** = Content
- **URL** = Naming/Location
- **HTTP** = Transfer Rules

The Evolution (Culture)

- **Web 1.0** = Static, Read-only delivery network.
- **Web 2.0** = Dynamic, Read/Write social platform powered by APIs and JS frameworks.